Hall Ticket No: | | | | | | | | | | |

Course Code: A8604

VARDHAMAN
COLLEGE OF ENGINEERING

III B.Tech I Semester Continuous Assessment – I, August - 2025

(Regulations: VCE-R22)

WEB TECHNOLOGIES

(Common to Information Technology & Computer Science and Engineering)

Date: 13 August 2025 FN **Time: 2 Hours** **Max. Marks: 30**

Answer all Questions in Part-A

Answer any FOUR Questions in Part-B

Course Outcomes with Bloom's Levels:

CO#	***CO Statement***	***Bloom's Level (L#)***
CO1	Design static and interactive web pages for a given application.	L3
CO2	Choose a web standard to store, transport and exchange in heterogeneous environments.	L3
CO3	Make use of JDBC API to interact with various databases.	L3
CO4	Deploy server-side components for providing services to the client.	L3
CO5	Design dynamic and data driven web pages in client-server environment.	L3

Questions:

			CO#	***BL#***	***Marks***
PART-A					
1.	a)	Use an HTML tag to insert an image with alternate text describing the image.	CO1	L2	***1 M***
	b)	Use a CSS selector to change the font color of all <h1> headings to red.	CO1	L2	***1 M***
	c)	Implement the margin property in CSS to add space around a paragraph of text.	CO1	L3	***1 M***
	d)	Write a html code to display table with 2 rows and 2 columns.	CO1	L2	***1 M***
	e)	Construct CSS for setting a background image.	CO1	L3	***1 M***
	f)	Demonstrate JavaScript code to validate an email input field.	CO2	L3	***1 M***
	g)	Apply JavaScript code to handle a button click event.	CO2	L3	***1 M***
	h)	Apply JavaScript code to calculate the sum of elements in an array.	CO2	L3	***1 M***
	i)	How can you apply a DTD to an XML document?	CO2	L3	***1 M***
	j)	Compare XML DTD and XSD?	CO2	L3	***1 M***
PART-B					
2.	Implement the form with various input types and demonstrate how it can be used to collect and submit data		CO1	L3	***5 M***

3.	Demonstrate CSS different mechanisms used to apply CSS for HTML pages?	CO1	L3	***5 M***
4.	Using the <svg> tag, create a simple scene that includes: • A rectangle representing the sky (light blue) • A yellow circle representing the sun • A green rectangle representing grass • At least one animate tag to animate the sun's position from left to right across the sky Write the full HTML and SVG code and explain how the animation works.	CO1	L4	***5 M***
5.	Examine different types of events in JavaScript by writing code that responds to user interactions on a web page. Show how to attach event handlers and manage events effectively."	CO2	L4	***5 M***
6.	Utilize three different JavaScript objects in a practical example. Demonstrate how each object is used to achieve specific tasks within a JavaScript program	CO2	L3	***5 M***
7.	Create DTD to define the structure of an XML document for a library system	CO2	L3	***5 M***

VARDHAMAN
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SCHEMA OF VALUATION

Name of the Subject	: Web Technologies	Subject Code	: A8604
Programme	: IT	Year	: III
Examination	: Regular / Supplementary	Date of Examination	: 13/08/2025
Max. Marks	: 30	Time of Examination	: FN.

Part - A

1. a) 1M

```
<img src = "logo.jpg" alt = "Vardhaman College
of Engineering" width = "500" height = "600">
```

b) 1M

```
<html>
<head>
<style> h1 { color : red ; } </style>
<body>
<h1> This is first h1 tag </h1>
<h2> This is h2 tag </h2>
<h1> This is Second h2 tag </h1>
</body> </head> </html>
```

c) 1M

```
<html>
<head> <style> p { margin : 20px ; } </style>
<body>
<p> This is my first paragraph </p>
<p> This is Web Technologies CAT I </p>
</body>
</html>
```

d) html table with 2 rows & 2 columns

```
<html>
<body>
<table border = "2" >
<tr> <td> S.No </td>
     <td> Roll no </td></tr>
<tr> <td> 1 </td>
     <td> 1201 </td></tr>
<tr> <td> 2 </td>
     <td> 1202 </td> </tr>
</table> </body></html>
```

S.No	Roll No
1	1201
2	1202

1M

e)

```
<html>
<head> <style>
  body { background-image : url ("c:/Downloads/logo.jpg");
         background-repeat : no-repeat ;
         background-position : center ;
       }
</style>
</head>
<body> This is css background image </body>
</html>
```

1M

f). Email regular Expression :-

/^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\\.[a-zA-Z]{2,}$/

Javasript

```
let regEx = /^[a-zA-Z0-9._%+-]+@
[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/
let Email = "test@Example.Com";
if(regx.test(Email)) {
Console.log("valid Email");
} else { Console.log("Invalid Email");
}
```

→ 1M

g) Student can take any example.

```
<html>
<button id="save_but"> Save </button>
<script>
    const btn = document.getElementById("save_btn");
    btn.addEventListener("click", () =>
          {console.log("Saved");});
</script>
```

→ 1M

h)

```
<script>
  const nums = [5,10,20,15,16];
   let sum=0;
   for(int i=0; i< nums.length ; i++)
   {
       Sum= sum + nums[i];
    }
   Console.log("Sum", Sum);
  document.write ("Sum =", +Sum);
</script>
```

→ 1M

(i)

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE bookstore [
    <!ELEMENT bookstore (book+)>
    <!ELEMENT book (title, author, price)>
    <!ELEMENT title (#PCDATA)>
    <!ELEMENT author (#PCDATA)>
    <!ELEMENT price (#PCDATA)>]
>.
```

dtd.

```
<bookstore>
  <book>
    <title> XML Guide </title>
    <author> John Smith </author>
    <price> 29.99 </price>
  </book>
</bookstore>
```

xml.

1M

(j)

xml + DTD	xml + xsd
(a) Uses its own special syntax, not written in XML	~~(a) Supports only basic types like <#P #PCDATA, CDATA, ID, etc>~~
(b) Supports only basic types like #PCDATA, CDATA, ID, etc	(a) written in xml format, so it can be printed and validated
	(b) supports rich built in types (str, int --)
(c) Does not supports xml namespaces	(c) fully supports xml namespaces for organizing elements & avoiding naming conflicts.

1M

PART-B

2) <html> ----

```
<body>
<h2>Sum of Numbers </h2>.
<form onsubmit = "Calculatesum(event)" >
<input type = "number", id= "n1" placeholder="Enter firstno" >
<input type = "number" id= "n2"                 "
<button type = "submit"> Calculate </button>
</form>
<p id = "result"></p>
<script>
function Calculatesum(event)
{
event.preventDefault();
let a = parseFloat (document.ElementbyId ("n1").value)
let b =      "        "          "n2"   "
document.getElementbyID ("result").InnerText="
Sum :"=
(a+b);
}.
</script>
</body>
</html>
```

Cd

→ 5M

3). Here is a single code snippet which covers all 3 CSS.

```
<!DOCTYPE html5>
<html>
  <head> <title> CSS Demonstration </title>
    <style>
      h2 { color : blue ; font-family : Arial, sans-serif ; }
      .internal-style { background-color : light yellow ; padding : 10px }
    </style>
    <!-- External >
      <link rel = "stylesheet" href = "style.css" >
  </head>
  <body>
    < h1 style = "color : red ; text-align : center ;" > Inline </h1>
    <p class = "inter-style" > Style using Internal css </p>.

    <!-- External CSS >
    <h3 class = " external-style > External CSS </h3>
    <p> Styled using a separate </p>
  </body>
</html>
```

Internal CSS

External CSS Linking

Inline

→ 3.5 M

External CSS

Save it with style.css

```
.external-style {
   color : green ;
   font-size : 20px ;
   border : 2px solid green ;
   padding : 5px ;
}
```

→ 1.5 M.

4)

```
<body>
  <h2> SVG Scene with animate Scen </h2>
  <svg width = "500" height" = "500">

  <rect x="0" y="0" width ="500" height" = "200" fill = "lightgreen"/>
                                                  → 1M
  <rect = x="0" y= "200" width = "500" hight = "100", fill = "green" />
                                                  → 1M
  <circle id = "sun" cx= "50" cy="60" r="30",
          fill = "yellow">
                                                  → 1M
  <animate attributeName = "cx" from= "50" to = "450".
      dur = "5s" repeatcount= "Indefinite" />
  </circle>                                       → 1M
  </svg>
```

Animate :

The <animate> tag moves the cx (center a position) from 50 to 450 continuously (repeatCount = "Indefinite").

→ 1M

5) - Click event (onclick)
- Mouseover event (mouseover)
- Keydown event (onkey down)
- Change event (onchange)

Explanation of any 3!

→ 3M

Ex:-
```
<body>
<button onclick="alert("Button clicked")"> Click </button>
<div
  onmouseover="this.style.background=="lightgreen""
  onmouseout= this.style.background=="lightgrey""
  style="width:100px; height=50px; background:lightgray;
  margin:10px;""> Hover Me </div>
<input type="text" onkeydown="console.log
("key pressed:" + event.key)" placeholder="Typehere">
</body>
</html>
```

→ 2M

6) Javascript Objects
(a) date object
(b) Math object
(c) Array object
(d) String object

Explanation any 3

→ 3M

VCE

Progsam :

```
<body>
  <h2> Javascipt Objects </h2>
  <p id = "Output"></p>
  <script>
    let outputText = " ";
    let now = new Date();
    outputText += "Current Date" + now.toLocalString() + "<br>";
    let randomNum = Math.floor(Math.random() * 100) + 1;
    let sqrtnum = Math.sqrt(randomNum).
    outputText += "Rand Num :" + randomnum + "<br>";
    "     = "Squar Rot :" + sqrtNum.toFixed(2);
    let fruits = ["Apple", "Banana", "Mango"];
    outputText += "fruits List :" + fruits.join(", ") + "<br>";
    outputText += "First :" + fruits[0] + "<br>";
    OutputText + = "Total " + fruits.Length;

    document.getElementById ("output").InnerHTML = outputText;
  </script>
</body>
```

Date Object

Math

Array Object

→ 2M

1) library.dtd.

```
<!ELEMENT library (book+)>
<!ELEMENT book (title, author, year genre)>
<!ATTLIST book id ID #REQUIRED>
<!ELEMENT title (#PCDATA)>
<!ELEMENT author (#PCDATA)>
<!ELEMENT year (#PCDATA)>
<!ELEMENT genre (#PCDATA)>
```

→ 2m.

library.xml

```
<?xml version="1.0" encoding="UTF-8">
<!DOCTYPE library SYSTEM= "library.dtd">
<library>
  <book id ="1001">
      <title>TreeColors</title>
      <author> Anvesh </author>
      <year> 2025 </year>
      <genre>Fiction</genre>
  </book>
  <book>
      ---
      ---
      ---
  </book>
</library>
```

→ 3m.